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6.0 INTRODUCTION

One of the main reference model for the data transmission is called open system interconnection (OSI) that defines seven layers. TCP/IP is another model that defines five layers.

We shall look at then layers for both OSI and TCP/IP model in table 1 and table 2 respectively.

Table 1 OSI Layer and Key features

OSI Layer	Key Features
Layer 1 Physical Layer	The layer establishes the physical connection between the devices and sends and receives the data over the wire.
Layer 2 Data Link layer	The layer links the devices
Layer 3 Network Layer	The layer is responsible for transmission of data from one device to another.
Layer 4 Transport layer	The layer is responsible for resequencing of data and retransmission if required.
Layer 5 Session Layer	The layer is responsible for establishing connection and authentication
Layer 6 Presentation layer	The layer is responsible for data formatting and encryption
Layer 7 Application Layer	The layer is responsible for running the application

TCP/IP model defines five layers as given in table 2.

TCP/IP Layer	Key Features
Layer 1 Physical Layer	The layer is responsible for physical connection
Layer 2 Data link layer	The layer is responsible to move the data frames
Layer 3 Network Layer	The layer is responsible for managing the packets that move across the network
Layer 4 Transport layer	The layer is responsible for establishing connection across application through various ports.
Layer 5 Application layer	The layer manages the applications that require network access

The TCP IP model information is communicated through packets. Each packet contains details such as their destination address and headers. TCP IP model each of the layers at the sender end add encoding and at the receiver and each of the layers decode the data.

Currently IPv4 is a popular standard protocol for the internet. The latest internet protocol version is IPv6 that uses 128 bit address. UDP (User Datagram Protocol) and ICMP (Internet Control Message Protocol) or other protocols used for data transmission. Udp is a stateless protocol where in the datagram does not precedes the previous one nor succeeds the next one. Udp is a popular protocol for streaming videos and gaming use cases. Icmp is a connectionless protocol mainly used for caring messages reporting errors.

In this unit we shall examine the core concepts of mobile IP networking.

6.1 OBJECTIVES

After going through this unit, you should be able to

- understand key concepts of mobile IP,
- understand various encapsulation and route optimization in IP Header,
- understand the concepts of mobility binding,
- understand the concepts of cellular IP,
- understand the mobile IP with IPV6,
- understand various concepts of IP Security.

6.2 MOBILE IP

The mobile IP allows users to communicate without using same IP address. The mobile IP also ensures that the user sessions are managed appropriately.

The main objective of the mobile IP is to ensure that mobile user's network connectivity is maintained when the mobile user changes location. Let us look at the common terminologies used in the mobile IP:

- Mobile Node (MN) is the mobile device such as smart phone that user has.
- Home Network defines the network which has the original assigned IP of the mobile node.

- Home Agent (HA) is the router of the home network to which the mobile node was originally connected to.
- Home Address indicates the permanent IP of the mobile node.
- Foreign Network defines the current network of the mobile node when the mobile user is on the move.
- Foreign Agent (FA) is the router in the foreign network to which the mobile node is currently connected to.
- Correspondent Node (CN) is the internet device that is connected to the mobile node
- Care-of Address (COA) represents the mobile node's temporary address when it is on the move.
- Foreign agent COA is the COA of the foreign agent.
- Co-located COA, the COA is co-located if the MN temporarily acquired an additional IP address which acts as COA. This address is now topologically correct, and the tunnel endpoint is at the MN. Co-located addresses can be acquired using services such as DHCP.

6.2.1 Need for Mobile IP

When a computer is shifted locations, the IP of the computer that is connected using TCP/IP protocol changes. The IP change is due to various factors such as change in the network, change of the ISP service provider, change of the network route tables, change in the network security and others.

Normally a mobile device is always on the move. If the IP of the mobile device changes, it impacts the connectivity and any ongoing communication. Hence mobile IP is required to ensure that the IP of the mobile device does not change.

6.2.2 Working of Mobile IP

Figure 1 depicts various components and the working of mobile IP

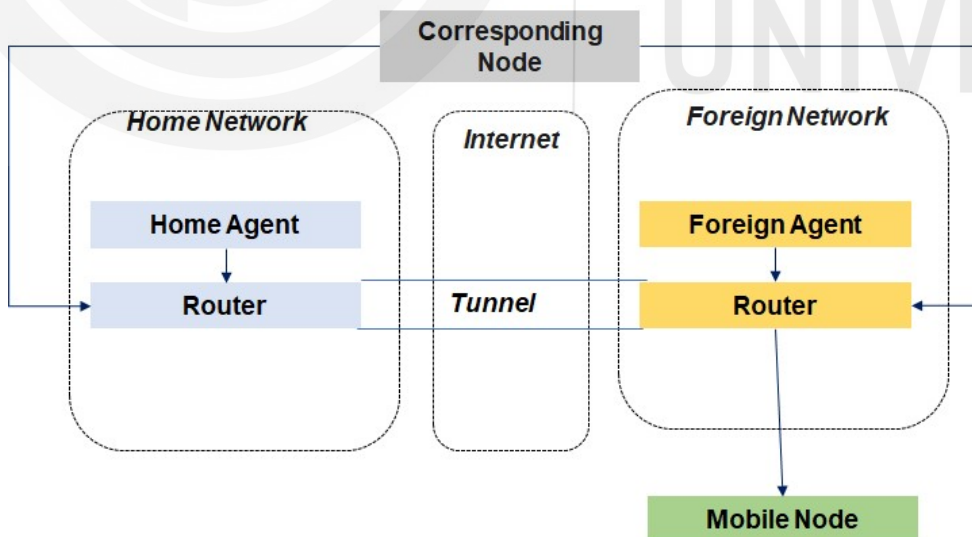


Figure 1: Working of Mobile IP

For the working of mobile IP, given below mechanisms are required:

- Agent advertising – The agents advertise their details on regular basis. The mobile node receives the details and identifies if it is a home network or a foreign network.
- Agent registration – The mobile node initiates the agent registration. If it is foreign agent, it sends a care of address to the home network after which the home agent replies to the foreign agent. The mobile node completes the agent registration.

Given below is the sequence of events for mobile IP:

1. The corresponding node sends the data packet to the home agent.
2. As the mobile node is not in the home network, the foreign network is used.
3. The foreign network sends the care-of-address of the mobile node.
4. A virtual pipe called tunnel is established over the internet between home agent and the foreign agent.
5. The home agent adds the care of address for the data packet and sends it to the foreign agent.
6. The foreign agent decapsulates the data packet and sends it to the mobile node.
7. The foreign agent sends the response from the mobile node to the corresponding node.

Check Your Progress – 1

1. _____ allows users to communicate without using same IP address.
2. _____ is the router of the home network to which the mobile node was originally connected to.
3. _____ is the internet device that is connected to the mobile node
4. _____ defines the network which has the original assigned IP of the mobile node.
5. _____ represents the mobile node's temporary address when it is on the move.
6. _____ is used over the internet for communication between home agent and the foreign agent.

6.3 IP HEADER: ENCAPSULATION AND ROUTES OPTIMIZATION

Let us look at the registration flow in greater details as depicted in Figure 2

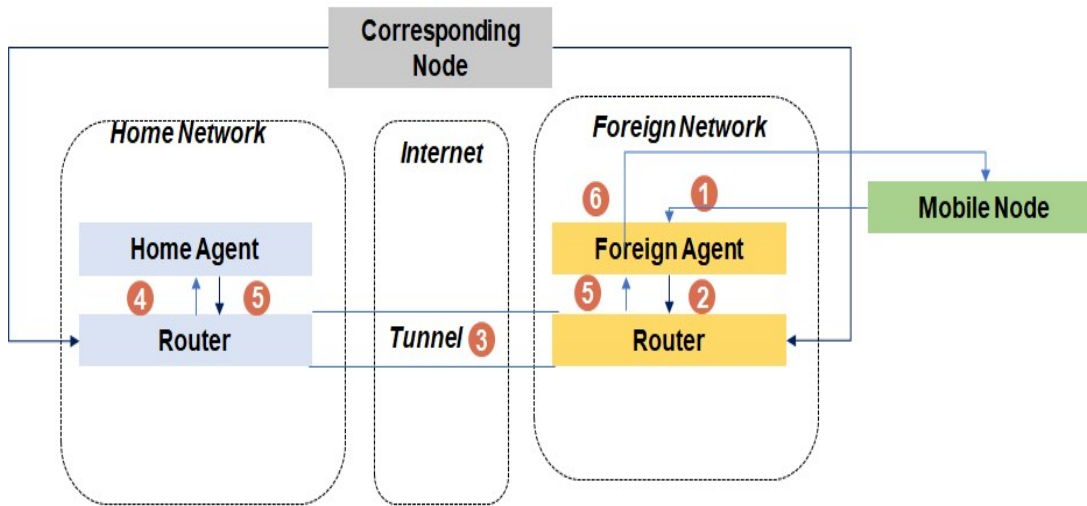


Figure 2: Agent registration Flow

Given below are the detailed steps involved in the agent registration:

1. The mobile node identifies the current network as a foreign network. The mobile node sends a registration request.
2. The foreign agent routes the request to the router
3. The router sends the care of address along with registration request through the tunnel. A tunnel is a virtual pipe that sends the data stream across two connected ends.
4. The router on the home network receives the request
5. The home agent sends the registration response.
6. The mobile node is registered and all the responses from the mobile node are sent to the home network.

6.3.1 Encapsulation of Additional IP Header

We have depicted the encapsulation of additional IP header in Figure 4.

Version	IHL	Service Type	Total Length	
Identification			Flags	Fragment offset
Time to live	Protocol		Header Checksum	
Source Address (Home Agent)				
Destination Address (Care of Agent)				

Figure 3: Additional IP Header encapsulation

The regular packet information mainly consist of Source and destination address and the IP header details. During the registration, the new encapsulated IP headers have below given details as shown in Figure 3:

- IP Version that depicts IPV4 along with header length and the total packet length.
- The flags and fragment offset details of the packet
- Time to live details that define the hops before the packet expiry, protocol and the header checksum
- The IP address of the source home agent.
- The IP address of the destination care of agent.

6.3.2 Minimum encapsulation by IP packet

The minimum encapsulation (ME) combines the data from the original packet as depicted in Figure 4.

Modified IP Header	Version	IHL	Service Type	Total Length	
	Identification			Flags	Fragment offset
	Time to live	Protocol		Header Checksum	
	Source Address (Home Agent)				
	Destination Address (Care of Agent)				
Minimal IP Header	Protocol	S	Reserved	Header Checksum	
	Destination Address (Original Sender)				
	Source Address (Home Address)				
	Payload				

Figure 4: Minimum Encapsulation

Given below are the changes in the Minimum encapsulation:

- The protocol type is specified
- Mobile Node IP address and corresponding node IP address is specified.

6.3.3 Generic Routing encapsulation by IP header

The existing IP headers described below does not provide the tunnel routing information, support recursive encapsulation and the authentication related key. Generic Routing Encapsulation (GRE) headers are added to address the above. The main elements of the GRE encapsulation are given below:

- The time to live flag is set to 1 that results in single forwarding.
- The GRE header is used for recursive encapsulation. The header consists of details such as recursion number, checksum, sequence number, routing information.

6.4 MOBILITY BINDING

We shall discuss the triangular route problem to understand the advantage of mobility binding.

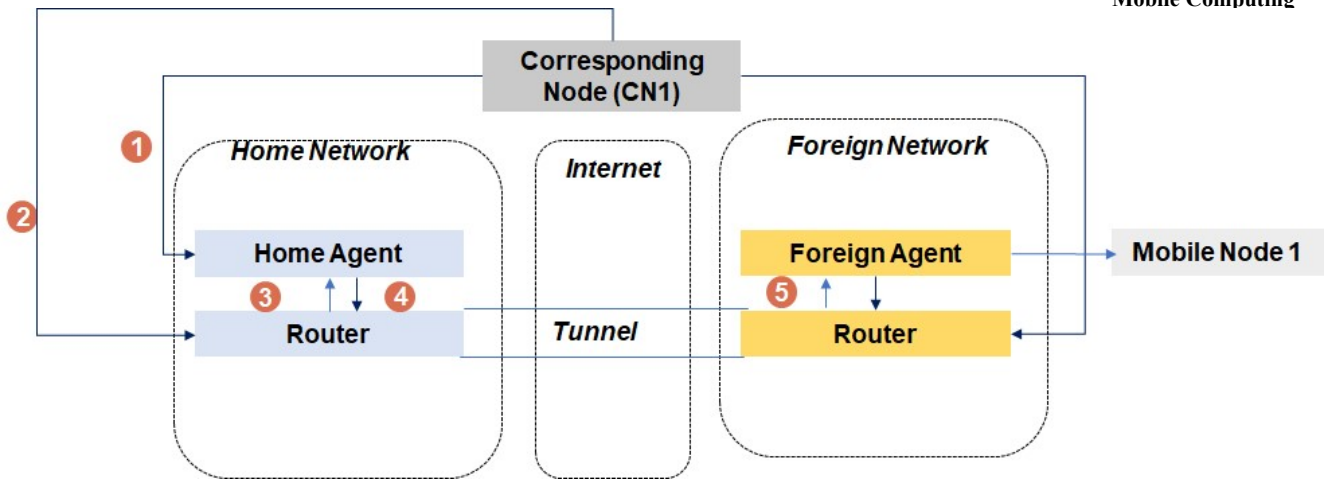


Figure 5: Triangular Routing

When the mobile node (MN1) visits a foreign network that is the home network of corresponding network CN2. Lets consider that CN2 is very close to the corresponding network CN1. For the CN1 to reach the MN1, it has to make a triangular trip visiting the home agent 1 and then reaching the COA at foreign agent as depicted in Figure 5.

The mobile binding optimizes this triangular routing by binding the CN1 directly to the COA of the MN1. This can be achieved using the mobility binding cache which caches the current COA of the Mobile node. The optimization achieved through Mobile binding is depicted in figure 6.

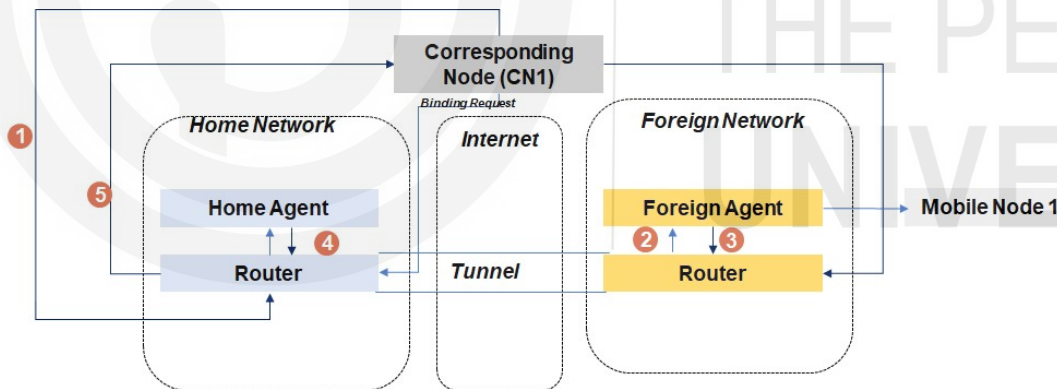


Figure 6: Route optimization through mobility binding

In figure 6 the CN1 directly sends the request to COA of the MN1 achieved through the mobility binding. Given below are the sequence of steps for mobility binding:

- The CN1 and MN1 request the HA1 for binding
- HA1 verifies the blocked status of MN1 and then sends the mobility binding message (consisting of MN1 IP and the current COA of MN1) for the CN1
- CN1 acknowledges the binding message to HA1.
- The binding is established MN1 and CN1.

6.5 CELLULAR IP

Mobile IP faces challenges such as handover duration, registration scalability. The cellular IP addresses these problems by complementing the mobile IP by providing fast and consistent handover control. Cellular IP facilitates connection across fast moving mobile objects within a cell.

Cellular IP provides following features:

- Cellular IP reuses existing packet formats, encapsulations
- Cellular IP is self-configuring
- Cellular IP provides fast handover within the cell and out of the cell.
- The cellular gateway connects to Internet through mobile IP

The high level architecture of cellular IP is depicted in Figure 7.

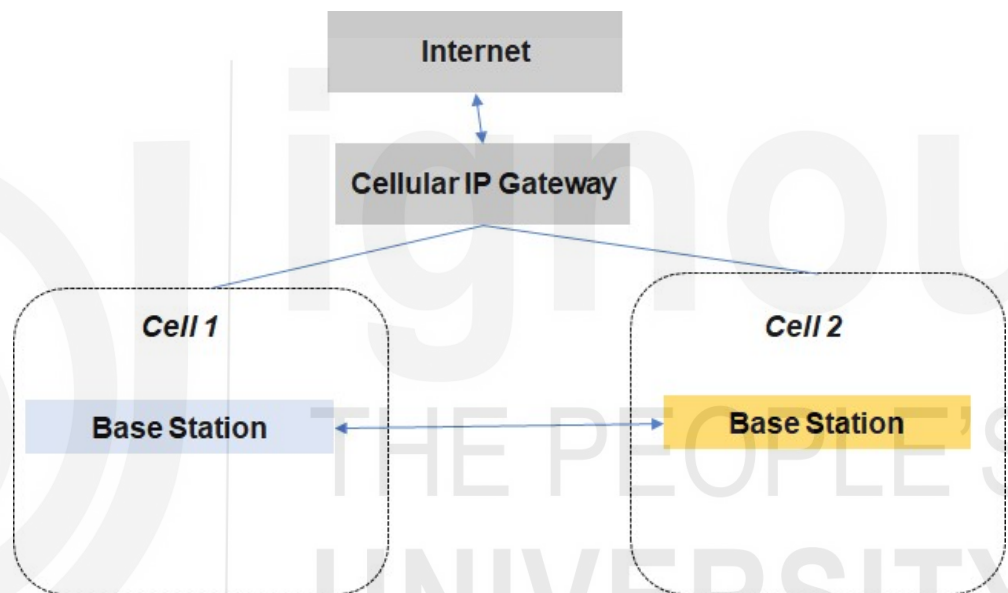


Figure 7: Cellular IP architecture

The cellular IP gateway filters the packets and the packets are routed to the corresponding home agent.

6.6 MOBILE IP WITH IPV6

IPV6 is the latest Internet protocol with 128-bit address. The IPV6 protocol specifies authentication header, routing, extension headers and others. Mobile IPV6 provides the mobility support for IPV6 providing a static IP address even when the mobile device changes location. Mobile IPV6 maintains connection across subnet changes.

Given below are the key features of mobile Ipv6:

- In Mobile IPV6, each mobile node has a static home address IP and the care-of-address IP.
- The home address IP is static and does not change when the location changes. The care-of-address IP switches to new network when the location is changed.
- The mobile node knows the IP of the home agent that makes the binding between home address and the received care of address.

- The home agent maintains a cache that stores the binding details of the home address and the care of address.
- The home agent forwards the data packet to the mobile nodes.
- The mobile node announces the care of address to the home agent.
- Mobile host binding happens through dynamic learning of mobile host binding
- The information packets are all authenticated.

6.7 VOICE OVER IP

Voice over IP (VoIP) facilitates the transfer of voice and multimedia over the Internet Protocol through Internet. VoIP facilitates local as well as long-distance calls. The VoIP enabled devices can send the audio voice, video call, fax and SMS over Internet. VoIP provides high reliability and smoother connection than the analog signals.

Figure 8 depicts a typical VoIP system. On the sender system, the voice and analog data is digitized and encoded at various layers.

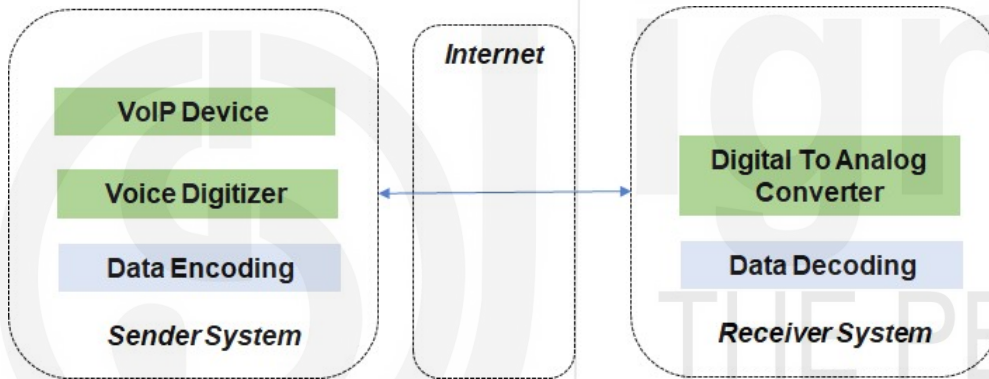


Figure 8: VoIP System

The data is then transmitted over the Internet (broadband, 4G networks and others) to the receiver system. On the receiver system, the data is decoded and digital to analog conversion is done to get the original voice signal.

There are various VoIP protocols such as session initiation protocol (SIP), mobile VoIP (mVoIP). SIP applications can be installed on mobile devices as they support multiple CODECs. mVoIP requires broadband or Wi-Fi networks.

Various modern applications such as Skype, MS Teams, Amazon Chime use the VoIP for audio and video conferencing.

6.8 IP SECURITY

IP security defines the set of protocols to secure the communication between two end points. IP security provides protocols for enforcing authentication, data confidentiality and data integrity. IP Security also defines the protocols for data encryption and data decryption and the associated secure key exchange.

IP Security is used for below given purposes:

- Encrypt the transmitted data.
- Establish the secure IPSec tunnels over the VPN.
- Provide the authentication for the transmitted data.
- Provide router security.

Let us now look at some of the key components of the IP security.

6.8.1 Key Components of IP Security

Encapsulating Security Payload (ESP)

The ESP provides various features such as encryption, data integrity validation and authentication payload. ESP also encrypts the payload data. The key ESP fields are header, encrypted data and integrity check value (ICV) that represents the ESP authentication data. ESP uses SH1 or MD5 for creating the ICV.

Authentication Header

We have depicted the main elements of authentication header in Figure 9. The authentication header provides the key features such as anti replay (prevents unauthorized packet transmission), data integrity and authentication. Algorithms such as SH1, MD5 are used for authentication.



Figure 9: Authentication Header

Security Association (SA)

The security association defines the shared security attributes and ensures the secure communication between two network entities. The SA also stores the packet encapsulation information including the keys, encryption algorithm, sequence number and others.

The security association uses security policy for packet filtering.

Internet Key Exchange (IKE)

IKE defines a secure protocol to dynamically exchange the encryption keys across two devices. IKE finds a security association between the two devices.

We can implement the message content protection by using a unique identifier for each of the packets. The identifier is used to determine the validity of the packet.

IPSec data encapsulation Modes

Transport mode and tunnel mode are the two data encapsulation modes of IPSec data. The transport mode consists of IP header, IPSec header and payload in layer 3. The IPSec tunnel mode provides IPSec header, new IP header over IP header and payload.

6.8.2 Working of IP Security

Given below are the detailed steps of IP Security:

1. Once the host decides the need for IPSecurity, the security policies are applied to the packets.
2. IKE Phase 1 kicks off wherein the communicating hosts authenticate themselves.
3. The security negotiation happens over the secure channel created in step 2.

4. IKE Phase 2 kicks off where the communicating hosts decide the cryptographic algorithms and secret key material used for the algorithms.
5. Exchange of data happens over the secure IPSec tunnel. The encryption and decryption happens through the IPSec SAs.
6. The tunnel is terminated after the completion of the communication or after timeout and the keys are discarded.

Check Your Progress – 2

1. In single tier architecture, mobile app interacts with the embedded database _____
2. _____ define the hops before the packet expiry, protocol and the header checksum in the IP Header
3. _____ is used for recursive encapsulation.
4. _____ optimizes this triangular routing by binding the CN1 directly to the COA of the MN1.
5. _____ compliments the mobile IP by providing fast and consistent handover control
6. _____ facilitates the transfer of voice and multimedia over the Internet Protocol through Internet
7. _____ defines the set of protocols to secure the communication between two end points

6.9 SUMMARY

OSI consists of 7 layers including Layer 1 Physical Layer, Layer 2 Data Link layer, Layer 3 Network Layer, Layer 4 Transport layer, Layer 5 Session Layer, Layer 6 Presentation layer and Layer 7 Application Layer. TCP/IP consists of 5 layers Layer 1 Physical Layer, Layer 2 Data link layer, Layer 3 Network Layer, Layer 4 Transport layer and Layer 5 Application layer. The mobile IP allows users to communicate without using same IP address. Mobile IP needs agent advertizing and agent registration. The tunnel between the Foreign agent and the home agent if the mobile device is in foreign network. The regular packet information mainly consist of Source and destination address and the IP header details. The minimum encapsulation (ME) combines the data from the original packet. Mobile binding can be used to optimize the triangular routing. The cellular ip complimenting the mobile IP by providing fast and consistent handover control. Voice over IP (VoIP) facilitates the transfer of voice and multimedia over the Internet Protocol through Internet. IP security defines the set of protocols to secure the communication between two end points. The ESP provides various features such as encryption, data integrity validation and authentication payload. The security association defines the shared security attributes and ensures the secure communication between two network entities.

6.10 SOLUTIONS/ANSWERS

Check Your Progress – 1

1. Mobile IP
2. Home Agent
3. Corresponding Node
4. Home Network

5. Care of Address

6. Tunnel

☛ **Check Your Progress – 2**

1. Time to live

2. GRE Header

3. Mobile binding

4. Cellular IP

5. Voice Over IP (VoIP)

6. IP Security

6.11 FURTHER READINGS

References

Mobile Computing 3rd Edition by Raj Kamal - <https://www.amazon.in/Mobile-Computing-Raj-Kamal/dp/0199455414>